TEST-POOL-01 (assessment test, not a real project)

TEST-DB-POOL

Customer	TEST customer
Job number	TEST-0001
Issued under	iec-default
Design rules	software defaults, not confirmed by the company's engineers

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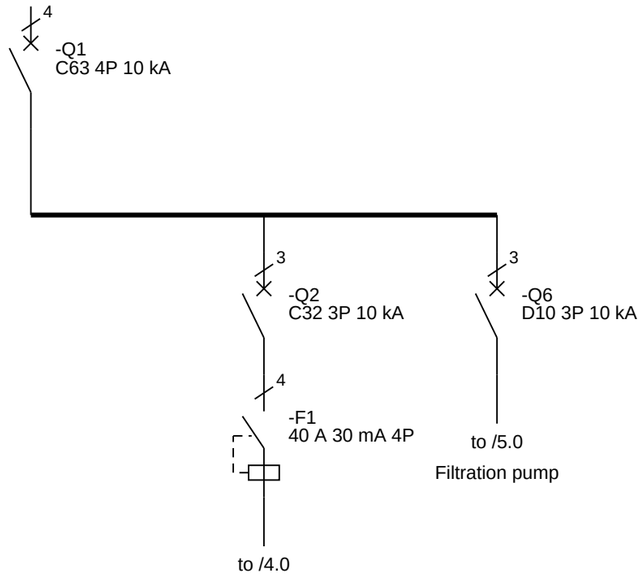
3/N/PE ~ 50 Hz
400 V TN-S
Ik 10 kA



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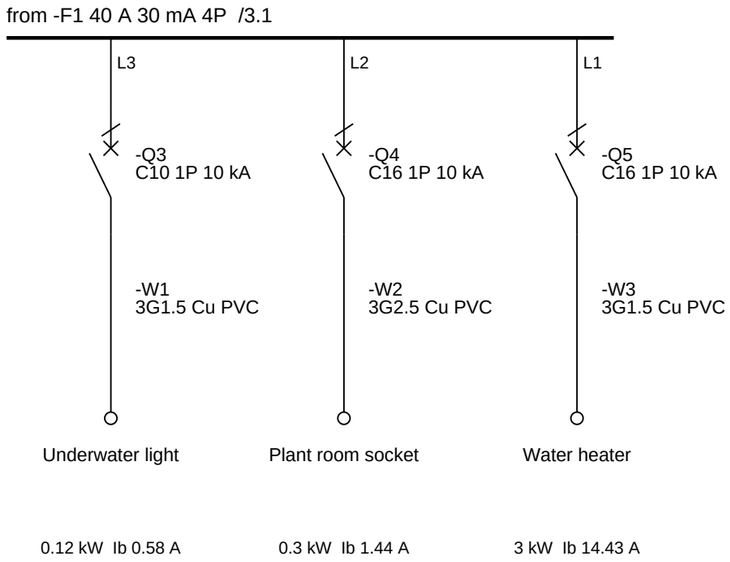

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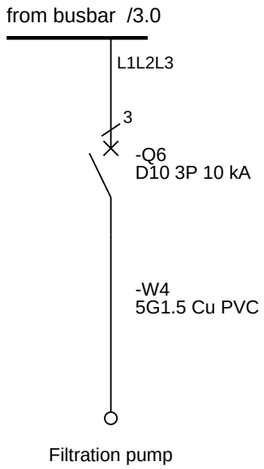

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No.	Note
1	Group 1 breaker raised to 32 A, 1.6 x the 16 A breaker after it, for discrimination.
2	Phase imbalance 59.6 % exceeds the company's 10 %: the loads cannot be spread more evenly as given.
3	Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination.
4	Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too.
5	Leave 1 spare outgoing ways (20 %).
6	Loads without a power factor were taken at cos phi 0.9.
7	Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.
8	4 circuit(s) have no cable length given; their voltage drop is not checked.
9	1 circuit(s) are not checked for earth fault disconnection: no cable length, a motor starter's breaker, or a breaker not selected here.
10	Each cable's withstand k^2S^2 is in the cable list; compare it with the selected breaker's let-through I^2t at 10 kA from its maker's curve (ABB handbook §2.4).
11	Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.
12	Designed for Palestine, under IEC 60364 (adopted by the Palestine Standards Institution): its stated limits apply beneath the company's own settings.
13	Circuit rules are this software's defaults, not confirmed by the company's engineers.
14	Enclosure: ABB Mini Center compact GDMS348 RX (2CVA360001P0001), 3 rows of 16 modules, H x W x D 648 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 31 modules are left for spare ways (ABB Distribution Boards catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted.

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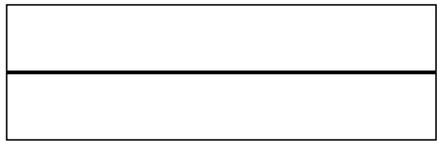
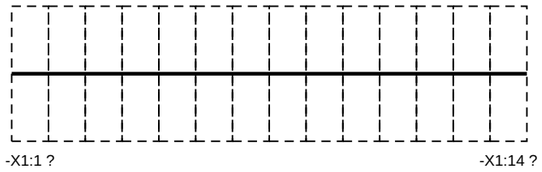
Incomer: 70 mm of rail



Residual current group: 175 mm of rail

Busbar: 52.5 mm of rail

Terminals: 0 mm of rail; 14 without a width



Dashed: no width is given for the device's kind (rail_widths_mm in the company settings).

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Terminal	Conductor	Circuit	Cable	mm²	Article
-X1:1	L3	Underwater light	-W1	1.5	8WH1000-0AF00
-X1:2	N	Underwater light	-W1	1.5	8WH1000-0AF00
-X1:3	PE	Underwater light	-W1	1.5	8WH1000-0CF07
-X1:4	L2	Plant room socket	-W2	2.5	8WH1000-0AF00
-X1:5	N	Plant room socket	-W2	2.5	8WH1000-0AF00
-X1:6	PE	Plant room socket	-W2	2.5	8WH1000-0CF07
-X1:7	L1	Water heater	-W3	1.5	8WH1000-0AF00
-X1:8	N	Water heater	-W3	1.5	8WH1000-0AF00
-X1:9	PE	Water heater	-W3	1.5	8WH1000-0CF07
-X1:10	L1	Filtration pump	-W4	1.5	8WH1000-0AF00
-X1:11	L2	Filtration pump	-W4	1.5	8WH1000-0AF00
-X1:12	L3	Filtration pump	-W4	1.5	8WH1000-0AF00
-X1:13	N	Filtration pump	-W4	1.5	8WH1000-0AF00
-X1:14	PE	Filtration pump	-W4	1.5	8WH1000-0CF07

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Cable	From	To	Type	Length
TEST-DB-POOL -W1	TEST-DB-POOL -Q3	Underwater light	3G1.5 Cu PVC	
TEST-DB-POOL -W2	TEST-DB-POOL -Q4	Plant room socket	3G2.5 Cu PVC	
TEST-DB-POOL -W3	TEST-DB-POOL -Q5	Water heater	3G1.5 Cu PVC	
TEST-DB-POOL -W4	TEST-DB-POOL -Q6	Filtration pump	5G1.5 Cu PVC	

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Qty	Description	Article	Order number	Devices
1	Circuit-breaker C63 4P 10 kA	not selected		-Q1
1	Circuit-breaker C32 3P 10 kA	not selected		-Q2
1	Residual current circuit-breaker 40 A 30 mA 4P	not selected		-F1
1	Circuit-breaker D10 3P 10 kA	not selected		-Q6
1	Circuit-breaker C10 1P 10 kA	not selected		-Q3
2	Circuit-breaker C16 1P 10 kA	not selected		-Q4, -Q5
10	Terminal 8WH1	Siemens 8WH1000-0AF00	8WH1000-0AF00	TEST-DB-POOL -X1
4	PE terminal 8WH1	Siemens 8WH1000-0CF07	8WH1000-0CF07	TEST-DB-POOL -X1

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